CSV to Parquet Data Processing

Dataset

Create a file named `employees.csv`

```
id,name,department,salary,experience
1,John,IT,50000,2
2,Alice,HR,45000,3
3,Bob,IT,70000,5
4,David,Finance,60000,4
5,Eva,IT,80000,7
6,Mary,HR,55000,6
```

Task 1: Read CSV (10 mins)

Read the CSV file and display:

* First 5 records
* Column names
* Number of rows and columns

Expected concepts:

* `pd.read_csv()`
* `head()`
* `columns`
* `shape`

Task 2: Data Analysis (10 mins)

Find:

1. Average salary
2. Maximum salary
3. Minimum salary
4. Number of employees in IT department

Expected concepts:

python
mean()
max()
min()

value_counts()

Task 3: Filtering (10 mins)

Display:

1. Employees with salary greater than 60000
2. Employees from IT department
3. Employees having experience greater than 4 years

Expected concepts:

```python
df[df["salary"] > 60000]
```

Task 4: Create New Column (10 mins)

Add a column:

text
salary_category

Rules:

* salary < 50000 → Low
* 50000 to 70000 → Medium
* > 70000 → High

Expected Output:

name	salary	salary_category
John	50000	Medium
Eva	80000	High

Task 5: Convert CSV to Parquet (10 mins)

Install:

```bash
pip install pyarrow
```

Convert CSV to Parquet.

```python
df.to_parquet("employees.parquet")
```

Task 6: Read Parquet and Verify (10 mins)

Read the Parquet file.

```python
df = pd.read_parquet("employees.parquet")
```

Display:

* Total rows
* Total columns
* First 3 records

Bonus Task (Advanced)

Group employees by department and calculate:

* Average salary
* Maximum salary

Expected Output:

```text
Department   Avg Salary
HR           50000
IT           66666
Finance      60000
```

Concept:

```python
df.groupby("department")["salary"].mean()
```

Here are 5 additional Parquet + Pandas tasks suitable for practice:

Task 1: Department-wise Employee Count

Read the Parquet file and display the number of employees in each department.

Expected Output:

```
IT          3
HR          2
Finance     1
```

Concepts:

value_counts()
groupby()
size()

Task 2: Top 3 Highest Paid Employees

Read the Parquet file and display the top 3 employees based on salary.

Expected Output:

```
Eva      80000
Bob      70000
David    60000
```

Concepts:

```
sort_values()
head()
```

Task 3: Update Salaries

Give a 10% salary hike to employees with more than 5 years of experience.

Create a new column:

updated_salary

Concepts:

loc[]
apply()

Task 4: Find Missing Values

Create a dataset with some missing values and:

1. Count missing values in each column.
2. Replace missing salary with average salary.
3. Save the cleaned data to a new Parquet file.

Concepts:

```
isnull()
fillna()
mean()
```

Task 5: Department Report Generation

Generate a summary report from the Parquet file:

Department	Employee Count	Avg Salary	Max Salary
IT	3	66666	80000
HR	2	50000	55000

Save the report as:

```
department_report.parquet
```

Concepts:

groupby()
agg()
to_parquet()

Challenge Task

Using the Parquet file, answer:

1. Who has the highest salary?
2. Which department has the highest average salary?
3. How many employees earn above the average salary?
4. What is the total salary expense of the company?
5. Which employee has the most experience?

This combines:

* Filtering
* Aggregation
* Sorting
* Parquet read/write
* Business reporting